AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

TM112
Introduction to Computing and Information Technology II

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Republic of Egypt

Mid-term Examination (MTA)
Spring Semester 2025/2026

Date: 16/April /2026

Number of Exam Pages: 3 *(including this cover sheet)*	**Time Allowed:** 2 hours

tructions:

- Total Marks: **30**
- This exam consists of **3 parts**.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are allowed.

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Part I: Multiple Choice Questions (5 marks).

1) The internet is
   A) A search engine
   B) A global network of networks
   C) A software application
   D) A social media platform
2) Convergence in smartphones refers to:
   A) Slower processing speed
   B) Merging of computing and telephone technologies
   C) Internet disconnection
   D) Smaller batteries
3) Which of the following is an example of biometric technology?
   A) Password
   B) PIN code
   C) Face recognition
   D) Username
4) Phishing attempts to:
   A) Speed up internet
   B) Protect your data
   C) Steal private information
   D) Update software

5) The largest unsigned number that can be stored in 3 bits is:
   A) 6
   B) 7
   C) 8
   D) 3

Part 2: Short questions (10 marks)

Question 1: (3 marks)

a) Differentiate between the internet and the web.
b) What is an information society?

Question 2: (4 marks)

For the given numbers find their equivalent in the specified base. **Show your steps.**

$(110111010)_2 = (\ )_{16}$
$(2AF)_{16} = (\ )_{10}$
$(3B4)_{16} = (\ )_8$
$(3.25*4)_{10} = (\ )_2$

Question 3: (3 Marks)

Define digital representation.
What is the difference between analogue and digital data?

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Part 3: Problem Solving / Long Questions (15 marks)

Question 1: (6 marks)

Consider an image of size **320 × 200** pixels.

Calculate how many **bits and bytes** are required to store this image in the computer:

a) Using black and white (2 colors).
b) Using 4 bits per pixel (grayscale).
c) Using RGB model.

Question 2: (5 marks)

a. What is usability?
b. What are the three components of usability?

Question 3: (4 marks)

A company wants to analyze millions of social media posts to detect customer sentiment.

a) What type of data is this?
b) What technology is suitable?

End of Que

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